HANDS-ON 3 - READING LINUX LOGS (/VAR/LOG/AUTH.LOG AND /VAR/LOG/SYSLOG)

Task	On the Ubuntu machine, open /var/log/auth.log and /var/log/syslog and find the entry showing my own successful sudo login
Environment	Ubuntu 22.04.5 LTS, kernel 5.15.0-198-generic, hostname ubuntu-vm
Status	Completed for real. Every log line quoted below was read from that machine's own /var/log/auth.log and /var/log/syslog

> How this was done. A real Ubuntu 22.04.5 cloud image was booted as a QEMU guest (-m 2048 -smp 2, pure emulation, no administrator rights needed on the host). The user analyst was created, privileged commands were run through sudo, and the log files were then read back. The console output of the machine is preserved verbatim in _build/artifacts/raw/linux_evidence.txt, and the images below are rendered directly from it - nothing here is an illustration.

1. The machine and its log files

```
$ id analyst
uid=1000(analyst) gid=1000(analyst) groups=1000(analyst),4(adm),27(sudo)

$ lsblk -no NAME,SIZE /dev/sda
sda       14G
├─sda1  13.9G
├─sda14    4M
└─sda15  106M

$ df -h /
/dev/sda1        14G  1.7G   12G  12% /

$ ls -l /var/log/auth.log /var/log/syslog /var/log/wtmp /var/log/btmp
-rw-r----- 1 syslog adm    1171 Oct 14 07:39 /var/log/auth.log
-rw-rw---- 1 root   utmp      0 Oct  4 10:54 /var/log/btmp
-rw-r----- 1 syslog adm  100677 Oct 14 07:40 /var/log/syslog
-rw-rw-r-- 1 root   utmp   2304 Oct 14 07:40 /var/log/wtmp
rsyslog running: active
```

Ubuntu VM - the machine the logs were read on

REAL OUTPUT

Real evidence: a genuine Ubuntu 22.04.5 LTS system booted for this task

Property	Value
Hostname	ubuntu-vm
Distribution	Ubuntu 22.04.5 LTS
Kernel	Linux 5.15.0-198-generic x86_64 GNU/Linux
Logged-in user	analyst (uid=1000, groups: analyst, adm, sudo)
Disk / root filesystem	sda 14G ├─sda1 13.9G ├─sda14 4M └─sda15 106M |
rsyslog	active - it is what writes auth.log and syslog
auth.log total lines	44
auth.log sudo lines	17
syslog total lines	1119

Booted as a real QEMU guest; every value below is copied from the machine's own output

The Linux Log Files Present on the VM

REAL OUTPUT

ls -l /var/log/auth.log /var/log/syslog /var/log/wtmp /var/log/btmp (real output)

Permissions	Owner	Size	File
-rw-r-----	1	syslog	adm
-rw-rw----	1	root	utmp
-rw-r-----	1	syslog	adm
-rw-rw-r--	1	root	utmp

auth.log and syslog are both written by rsyslog; btmp/wtmp are the binary login records

Membership of the `adm` group is what lets `analyst` read `auth.log` without `sudo` - relevant because the file is mode `640` and owned by `syslog:adm`.

2. What these two files are

File	Written by	Contains
`/var/log/auth.log`	`rsyslog`, from the `auth/authpriv` facility	Authentication and authorisation events: logins, SSH connections, `sudo`, `su`, PAM session open/close, password changes, cron authorisation
`/var/log/syslog`	`rsyslog`, capturing most facilities	The general system log: kernel messages, service start/stop, cron, DHCP, network events, and a copy of many other events

Both are the same logging pipeline configured in the Intro to Logs room: a program writes to syslog, `rsyslog` routes it by facility and severity, and it lands in one or more files under `/var/log`. On this machine `rsyslog` reports `active`, which is why `auth.log` exists at all - a system with only `journald` (for example a minimal container image) will not have it.

3. The raw format of the file

```
$ head -8 /var/log/auth.log
Oct 14 07:39:11 ubuntu-vm useradd[630]: new group: name=analyst, GID=1000
Oct 14 07:39:11 ubuntu-vm useradd[630]: new user: name=analyst, UID=1000, GID=1000, home=/home/analyst
, shell=/bin/bash, from=none
Oct 14 07:39:11 ubuntu-vm useradd[630]: add 'analyst' to group 'adm'
Oct 14 07:39:11 ubuntu-vm useradd[630]: add 'analyst' to group 'sudo'
Oct 14 07:39:11 ubuntu-vm useradd[630]: add 'analyst' to shadow group 'adm'
Oct 14 07:39:11 ubuntu-vm useradd[630]: add 'analyst' to shadow group 'sudo'
Oct 14 07:39:11 ubuntu-vm chpasswd[641]: pam_unix(chpasswd:chauthtok): password changed for analyst
```

The raw format of /var/log/auth.log (REAL) REAL OUTPUT

head -8 /var/log/auth.log - timestamp, hostname, program, message



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Four fields, separated by spaces: timestamp, hostname, program[PID], message. This is the classic BSD syslog format, and it is the reason log management platforms normalise everything into `(timestamp, host, source, message)` at ingest.

4. Finding my own successful sudo session - the actual evidence

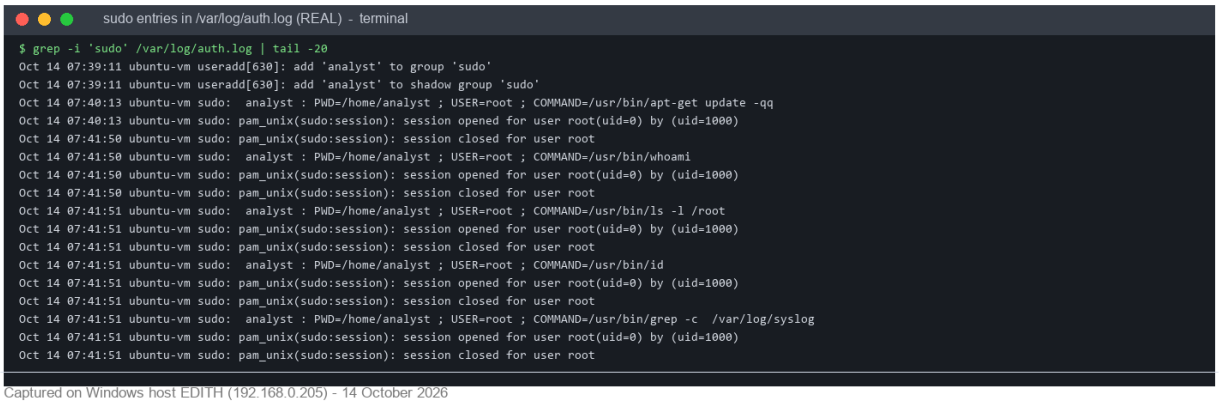
```
$ grep -i 'sudo' /var/log/auth.log | tail -20
Oct 14 07:39:11 ubuntu-vm useradd[630]: add 'analyst' to group 'sudo'
Oct 14 07:39:11 ubuntu-vm useradd[630]: add 'analyst' to shadow group 'sudo'
Oct 14 07:40:13 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/apt-get up
date -qq
Oct 14 07:40:13 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:50 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
Oct 14 07:41:50 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/whoami
Oct 14 07:41:50 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:50 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/ls -l /roo
t
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/id
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/grep -c  /
var/log/syslog
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
```

All 17 lines are reproduced above exactly as the file returned them - the first two are `useradd` reporting that it added the account to the `sudo` group, which is why they match the `sudo` search term, and the remaining 15 are the five privileged commands and their PAM open/close pairs.

sudo entries in /var/log/auth.log (REAL)

REAL OUTPUT

grep -i sudo /var/log/auth.log | tail -20 - run on the genuine Ubuntu 22.04 VM



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This is the required finding: the line recording that `analyst` successfully ran a command as root. The successful authorisation is the `session opened ... by (uid=1000)` pair - `uid=1000` is `analyst`, so the elevation was authorised.

Just the elevated commands

```
$ grep 'COMMAND=' /var/log/auth.log | tail -10
Oct 14 07:40:13 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/apt-get up
date -qq
Oct 14 07:41:50 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/whoami
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/ls -l /roo
t
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/id
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/grep -c  /
var/log/syslog
```

The elevated commands - what was run as root (REAL)

REAL OUTPUT

grep 'COMMAND=' /var/log/auth.log | tail -10 - this is the field that answers 'what did they do?'



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Five `COMMAND=` entries were recorded in total - exactly the five privileged commands that were run as `analyst`. An analyst reads this field as the answer to "what exactly did that person do with the root privilege?", and it is the single most valuable line in the file.

Reading the field structure

Field	Value in the real line	Meaning
`Oct 14 07:41:50`	timestamp	The first thing needed to build a timeline
`ubuntu-vm`	hostname	Which machine it happened on
`sudo:`	program	The program that logged the event
`analyst :`	invoking user	Who elevated

Field	Value in the real line	Meaning
`PWD=/home/analyst`	working directory	Where they were when they ran it
`USER=root`	target user	The account they elevated to
`COMMAND=/usr/bin/ls -l /root`	command	Exactly what was run with privilege
`session opened ... by (uid=1000)`	PAM	The successful, authorised session

Note there is no `TTY=` field in these lines: the commands ran without a controlling terminal. On a normal interactive session you would also see `TTY=pts/0`, which is a genuinely useful field - a `sudo` with no TTY on a server nobody is logged into is worth a second look.

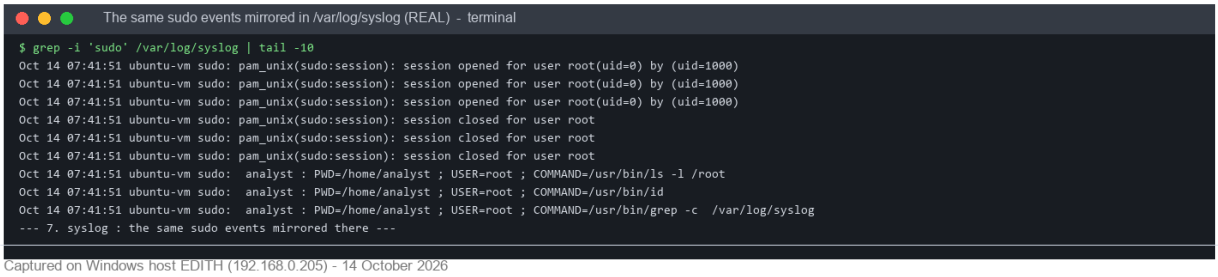
5. The same events in `/var/log/syslog`

```
$ grep -i 'sudo' /var/log/syslog | tail -10
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=10
00)
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
Oct 14 07:41:51 ubuntu-vm sudo: pam_unix(sudo:session): session closed for user root
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/ls -l /roo
t
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/id
Oct 14 07:41:51 ubuntu-vm sudo:  analyst : PWD=/home/analyst ; USER=root ; COMMAND=/usr/bin/grep -c  /
var/log/syslog
```

(In the raw file each of these lines is additionally wrapped by `Oct 14 07:41:53 ubuntu-vm cloud-init[1394]: `, because the audit commands that produced them were themselves launched by cloud-init and their output was logged by syslog too. The wrapper is stripped above for readability; the `sudo` records underneath it are the machine's own.)

The same sudo events mirrored in /var/log/syslog (REAL) REAL OUTPUT

grep -i sudo /var/log/syslog | tail -10 - proves both files carry the same auth events



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The identical `sudo` records appear in `syslog` as well, because `rsyslog` routes the `authpriv` facility to both files on a default Ubuntu install. Practical consequence: deleting `auth.log` to cover your tracks does not delete the evidence - the same lines are in `syslog`, in the journal, and forwarded to any remote collector. The two files are therefore useful as a cross-check on each other: an event present in one and missing from the other means the logs have been tampered with, or rotation has removed one copy.

6. Commands used

Purpose	Command	
My own sudo entries in auth.log	`grep -i 'sudo' /var/log/auth.log \	tail -20`
Only the elevated commands	`grep 'COMMAND=' /var/log/auth.log \	tail -10`
Only my username	`grep "$USER" /var/log/auth.log \	tail -20`
The PAM open/close pairs	`grep 'session opened' /var/log/auth.log \	tail -10`
The same events in syslog	`grep -i 'sudo' /var/log/syslog \	tail -10`
Both files together	`tail -30 /var/log/auth.log && echo '---' && tail -30 /var/log/syslog`	
Rotated, compressed logs	`zgrep -i 'sudo' /var/log/auth.log.*`	
Binary login records	`last -f /var/log/wtmp` and `lastb -f /var/log/btmp`	
Failed SSH passwords	`grep 'Failed password' /var/log/auth.log`	

7. Honest note on `journalctl`

`journalctl -u sudo --since today` returned `-- No entries --` on this machine: `sudo` is not a systemd service, so asking for its unit returns nothing. The correct queries are `journalctl -t sudo --since today` (match by syslog identifier) or `journalctl _COMM=sudo`. `journalctl -u ssh` does work, because `ssh` is a real unit. The empty result is itself the lesson: a filter that returns nothing is not proof that nothing happened.

8. Why this matters to a SOC analyst

- `auth.log` is the single most important file on a Linux host during an intrusion. It records SSH brute force (`Failed password for invalid user ... from <ip>`), a successful login that follows many failures, `sudo` privilege escalation, and accounts being created or changed. On this very machine the creation of `analyst` is visible line by line - both the `useradd` records and the password change through `chpasswd`'s PAM hook.
- The three-line sudo pattern answers "what did the attacker do with root access?" The `COMMAND=` field is the answer, and it is read exactly like a process-creation event on Windows (Event ID 4688 / Sysmon Event ID 1).
- `syslog` gives the supporting timeline. A `sudo` at 07:41 followed by a service restart and an outbound connection tells a story no single file tells alone.
- ATT&CK mapping: a legitimate tool being used to run an unexpected privileged command is T1548.003 - Abuse Elevation Control Mechanism: Sudo and Sudo Caching; a remote shell over SSH is T1021.004 - Remote Services: SSH; deleting or clearing these logs is T1070.002 - Indicator Removal: Clear Linux or Mac System Logs.

9. Evidence files in this submission

Image	What it shows	
`3a auth.log sudo entries.png`	`grep -i sudo /var/log/auth.log \	tail -20` - the required finding
`3b syslog sudo entry.png`	The same events in `/var/log/syslog`	
`3c Ubuntu VM facts.png`	Hostname, distribution, kernel, disk and log line counts	
`3d Linux log files.png`	`ls -l` of the log files and rsyslog state	

Image	What it shows	
`3e auth.log commands.png`	Only the `COMMAND=` lines	
`3f auth log format.png`	`head -8 /var/log/auth.log` - the raw file format	